A Physics-Guided Reconstruction and Injection Framework With Semantic Consistency Constraints for Oriented SAR Target Detection

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***Abstract*—Synthetic aperture radar (SAR) target detection is pivotal for a wide range of applications, including maritime surveillance, reconnaissance, and strategic security. However, complex scattering mechanisms often render SAR targets discrete and discontinuous, leading to severe missed detections and false alarms. To address this, this article proposes the physics-guided reconstruction and injection framework (PRIF), which introduces the attributed scattering center (ASC) model to characterize these discrete physical scattering properties. The framework comprises the topology-guided semantic consistency matching (TSCM) module, the part-aware label assignment (PALA) strategy, and the physics-guided reconstruction and injection network (PRI-Net). Specifically, the TSCM module ensures the semantic consistency of physical parts by resolving the label permutation problem in clustering, while the PALA strategy effectively mitigates the scattering voids issue by achieving spatial alignment between sampling points and physical entities. The core architecture, PRI-Net, effectively boosts part-level local feature discrimination and significantly improves model interpretability. It achieves this by internalizing physical priors into feature representations through the physics-guided progressive reconstruction branch (PPRB) and performing explicit feature injection via the decoupled spatial attention mechanism of the physics-injected dual-drive detection branch (PDDB). The experimental results on the FAIR-CSAR dataset demonstrate that PRIF achieves an mAP50 of 0.468 and an *F*1-score of 0.547, significantly outperforming existing state-of-the-art detectors. Furthermore, cross-dataset validation on the RSAR dataset and portability experiments on baseline models demonstrate its robust generalization capabilities and plug-and-play characteristics.**

***Index Terms*—Attributed scattering center (ASC), physics-guided learning, semantic consistency, synthetic aperture radar (SAR) image interpretation, SAR target detection.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is a sophisticated active microwave remote sensing modality that has established itself as a cornerstone in ocean surveillance, disaster assessment, and military reconnaissance [1], [2], [3], [4]. This prominence is primarily due to its inherent all-weather, all-day, and high-penetration observation capabilities [5], [6], [7]. Among numerous application scenarios, SAR aircraft target detection and recognition hold irreplaceable value for airport scheduling management and strategic intelligence analysis.

The proliferation of deep learning within computer vision has propelled convolutional neural networks to the forefront of SAR image interpretation, markedly enhancing automated processing capabilities [8], [9], [10]. Faster R-CNN innovatively proposed the region proposal network, establishing the classic paradigm of two-stage detection [11]; while RetinaNet effectively solved the positive–negative sample imbalance problem in one-stage detection through focal loss [12], becoming a representative architecture for efficient detection. FCOS discarded complex anchor box designs and simplified the model architecture through a point-based fully convolutional prediction mechanism, enhancing adaptability to scale variations [13].

Recently, DETR revolutionized the design paradigm of traditional detectors by introducing the Transformer architecture and leveraging self-attention to model long-range contextual relationships [14]. Aiming at arbitrarily oriented targets, researchers have made specific improvements based on the aforementioned general frameworks: methods such as S2ANet [15] and R3Det [16] effectively alleviated the spatial misalignment problem between rotated target features and horizontal convolution kernels by introducing feature alignment mechanisms. Meanwhile, to mitigate boundary discontinuities in angular regression, GWD [17] and KLD [18] represent rotated bounding boxes as 2-D Gaussian distributions, thereby streamlining the optimization of oriented objects.

Despite their impressive performance, these detection frameworks rely predominantly on a data-driven paradigm [19]. Consequently, these methods often neglect the fundamental SAR imaging physics and the intrinsic scattering mechanisms [20], [21], [22].

SAR aircraft targets are fundamentally different from optical targets, exhibiting significant characteristics of discrete scattering [23], [24], [25], multicenter [26], [27], and scattering voids. Specifically, the wings, engines, and tail edges

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form intense scattering points due to corner reflection effects, whereas the smooth fuselage surface produces specular reflection that results in a lack of echoes, thereby creating scattering voids at the geometric center of the image. These physical characteristics cause target features to be highly discrete in space [28], [29], making it difficult for general algorithms based on visual texture and geometric center priors to capture the overall structure of the target, easily leading to false responses or localization drift [30], [31].

To compensate for the lack of physical information, some researchers have attempted to introduce attention to electromagnetic scattering characteristic information [32], [33], [34], [35]. Meng et al. [23] proposed a scattering topology cue-based network, utilizing an improved Harris–Laplace detector to extract strong scattering points combined with DBSCAN clustering and Gaussian mixture models to enhance the scattering topology feature representation of aircraft targets in SAR images. Guo et al. [36] introduced a detection framework that synergizes scattering information enhancement with an attention-based pyramid architecture to address SAR target variability. Huang et al. [37] proposed a physics-guided aircraft detector, integrating the electromagnetic scattering physical mechanism into the network design and using scattering center distribution as a prior constraint to improve robustness in complex scenes. However, most of these methods only utilize Amplitude data, ignoring the potential massive scattering characteristics contained in the phase information of SAR complex data, which limits the model's deep understanding of the target's physical essence.

To fully mine the physical information in phase data, the attributed scattering center (ASC) part model is considered an ideal bridge connecting the physical world and feature space, because it can effectively parameterize the physical geometric structure of targets [38]. This idea aligns with the academic paradigm of hybrid modeling [19], which is dedicated to organically combining SAR physical models or physical prior information with deep neural networks. In recent years, exploration in this direction has mainly focused on two core dimensions: physical model representation based on feature fusion and physics-inspired model architecture design.

The first dimension involves physical model representation based on feature fusion, focusing on the transformation of abstract ASC parameters into efficient feature forms. These representations are designed to interact effectively with neural networks, thereby achieving a complementary enhancement between physical features and deep semantic features. Zhang et al. [39] proposed a feature extraction framework based on the bag-of-words model, which augments CNN features by incorporating quantized ASC feature vectors as physical supplements. Liu et al. [40] utilized geometric deep learning concepts to treat the ASC parameter set as a 3-D point cloud and extracted its spatial topological features through an improved point cloud network. Feng et al. [41] further exploited the reversibility of the ASC model to inversely reconstruct parameters into noise-free part-level images, achieving pixel-level alignment between physical and visual features. On this basis, to address the compatibility issue of heterogeneous features, Zhang et al. [39] introduced discriminant correlation analysis to maximize the correlation between modalities and eliminate redundancy.

The second dimension is physics-inspired or theory-guided model architecture design. Scholars have attempted to directly embed electromagnetic physical knowledge into network architectures. Specifically, scattering center knowledge is explicitly integrated into the feature representation layer to enhance the model's perception of underlying physical mechanisms. For instance, Feng et al. [42] designed the electromagnetic scattering feature embedding network, which explicitly embeds physical priors into the feature extraction layers. In this direction, Huang et al. [19] proposed a physics-inspired hybrid attention mechanism. They adaptively integrated the physical priors of the ASC into the spatial and channel feature reweighting process of deep networks, thereby significantly improving the model's robustness and physical interpretability under complex operating conditions.

However, it must be noted that whether it is cascade fusion based on the feature level or physics-guided design based on the architecture level, the research focus has mainly been on the field of single-target recognition [19], [41], [42]. These ASC part models have not been effectively applied to the more challenging task of multitarget detection and recognition. Especially in scenarios with complex background interference and multitarget coexistence, the aforementioned hybrid modeling methods are often difficult to transfer directly. First, the background information of multiple targets in detection scenarios is extremely complex and difficult to remove, directly leading to the failure of refined ASC parameter extraction. Second, stitched images of multitarget parts lack semantic consistency; a single part image may mix parts from different aircraft, such as wings and nose, making it difficult to unify the modeling of information representing different semantics. Finally, traditional ASC part models strictly rely on complex data not only for training but also for parameter extraction and network integration during inference. This dependency constitutes a formidable data threshold for practical applications, where often only amplitude-only images are available.

Although a few existing studies have innovatively introduced the ASC model into the field of SAR target detection and recognition [43], they have not fully mined the geometric topology and high-level semantics of parts. Moreover, they are limited to horizontal box detection, making them unable to cope with the complex challenges of oriented detection and recognition of SAR targets.

To address the disconnect between data-driven approaches and physical scattering mechanisms, this article proposes the physics-guided reconstruction and injection framework (PRIF). First, to address the semantic misalignment problem of ASC clustering, we constructed a topology-guided semantic consistency matching (TSCM) module. TSCM utilizes the Hungarian algorithm to perform global optimal matching between disordered physical scattering points and ordered semantic anchors, converting discrete scattering points into semantically aligned supervisory signals.

Second, to internalize physical priors as inherent attributes of feature representations, this article proposes a physics-guided reconstruction and injection network (PRI-Net),

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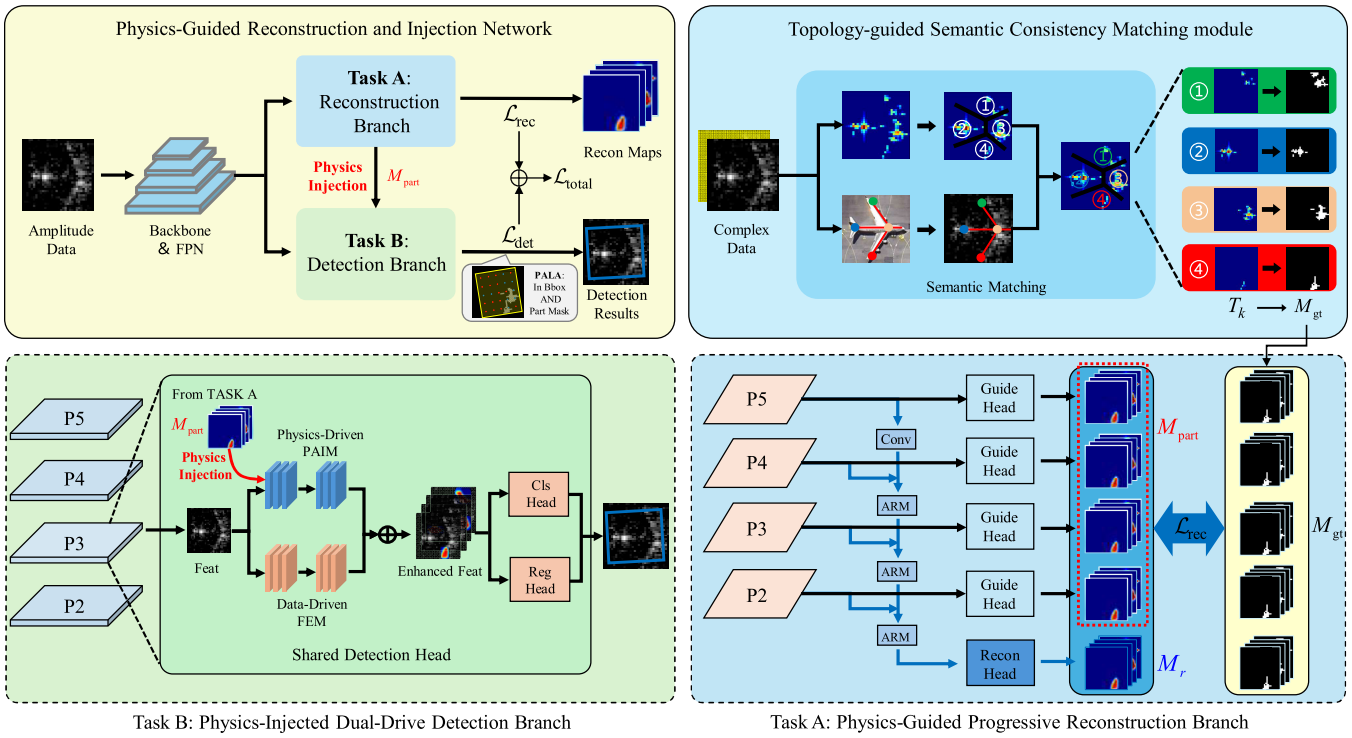


Fig. 1. Overall framework of PRIF. The PRIF comprises the TSCM module, the PALA strategy, and the core PRI-Net. Within the core network, the PPRB and the PDDB collaborate to achieve the deconstruction of physical priors and the explicit injection of features.

comprising a physics-guided progressive reconstruction branch (PPRB) and a physics-injected dual-drive detection branch (PDDB). PRI-Net constructs an auxiliary physical reconstruction branch to force the network to recover complete geometric topological structures from discrete scattering features. It explicitly injects physical features into specific channel groups of feature maps via a decoupled spatial attention mechanism and adopts a strategy of cross-domain supervision and single-domain inference: relying on complex data to generate mask supervision only during the training phase while requiring only amplitude images during the inference phase. Finally, to address the discrete nature of SAR targets, we propose the part-aware label assignment (PALA) strategy, abandoning geometric center sampling in favor of positive sample allocation based on the principle of physical entity alignment. The core contributions of this work are threefold as follows.

1) We propose the PRIF and its core network, PRI-Net, constructing a collaborative framework that harmonizes physical priors with deep data perception. The PRI-Net internalizes physical priors into feature representations through the PPRB and achieves explicit part-level semantic recalibration rather than coarse feature stacking. This not only avoids the dilution of local physical sensitivity by global contexts but also plays a robust global spatial filtering role, significantly enhancing the capability of the detector to suppress complex background clutter.
2) We propose the TSCM module, which elevates traditional pixel-level or holistic mask reconstruction to multidimensional physical semantic decoupling. By introducing constraints on part semantic consistency, it effectively solves the label permutation problem in unsupervised ASC clustering. This module transforms disordered physical scattering centers into topologically ordered structural penalty terms, thereby establishing an effective mapping bridge between underlying electromagnetic scattering features and high-level semantic concepts.
3) We design the PALA strategy specifically to address the scattering voids problem inherent in discrete SAR aircraft targets. PALA abandons traditional geometric center sampling criteria and establishes a positive sample allocation principle based on physical entity alignment. By strictly anchoring spatial sampling points to the semantically aligned features in TSCM, it significantly improves the capture capability and localization precision for weak discrete targets.

II. Methodology

A. Overall Framework of PRIF

This article proposes a PRIF based on the ASC model, which integrates part semantic consistency constraints for SAR aircraft target detection and recognition, as shown in Fig. 1. Most existing paradigms originate from the traditional computer vision field, primarily targeting optical images with continuous textures and dense features. In contrast, our PRIF framework is specifically designed for the unique electromagnetic scattering characteristics of SAR images. Inspired by FCOS [13], our framework leverages its anchor-free mechanism to flexibly adapt to the multiscale characteristics of SAR targets. Furthermore, the decoupled pixelwise prediction architecture provides an optimal structural foundation for

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embedding the physical reconstruction branch and implementing the discrete sampling strategy.

To address the refined part reconstruction requirements proposed in this article, we adaptively reconstructed the feature pyramid network (FPN): the feature output is adjusted to a four-layer structure from P_2 to P_5, which represent the feature maps at four different spatial scales. By introducing the high-resolution shallow features of the P_2 layer, the network can more keenly capture minute scattering point details, which aids in the high-quality recovery of the target's part-level geometric morphology.

The image is first input into the ResNet50 backbone for feature extraction. The generated feature maps are fused through the improved four-layer FPN, outputting a feature pyramid set, denoted as $\mathcal{P} = \{P_2, P_3, P_4, P_5\}$. Subsequently, $\mathcal{P}$ is fed into two parallel task branches: Task A is the reconstruction task and Task B is the detection task.

To provide precise supervisory signals with semantic consistency for the part-level reconstruction task, this article preprocesses the complex data I_{com} via the TSCM module. Furthermore, this article employs the PALA strategy to strictly correct the sampling criterion to physical entity alignment.

Specific innovative designs and implementation details regarding core modules such as physical injection, semantic consistency, and label assignment are elaborated in Sections II-C–II-E.

B. Target Part Extraction Based on ASC Model

To deconstruct target components with clear physical meanings from complex SAR data, this study uses the attribute scattering center model as the basis for parametric representation. Based on the geometrical theory of diffraction under high-frequency approximation conditions, the electromagnetic backscattering field of complex targets like aircraft is not continuously distributed. Instead, it can be modeled as the coherent superposition of several discrete and independent scattering centers. According to the classic parameterized model established by Gerry et al. [44], each scattering center not only contains location information but also describes its geometric attributes and scattering mechanism through a set of specific physical parameters. Therefore, the target backscattering response $E(f, \varphi; \Theta)$ can be mathematically represented as the summation of the response functions of multiple independent scattering centers

$$E(f, \varphi; \Theta) = \sum_{i=1}^{q} E_i(f, \varphi; \theta_i) \tag{1}$$

$$\begin{aligned} E_i(f, \varphi; \theta_i) = A_i \cdot \left(\mathrm{j}\frac{f}{f_c}\right)^{\alpha_i} \cdot \exp\left(-2\pi f \gamma_i \sin\varphi\right) \\ \cdot \exp\left(-\mathrm{j}\frac{4\pi f}{C}\left(x_i \cos\varphi + y_i \sin\varphi\right)\right) \\ \cdot \operatorname{sinc}\left(\frac{2\pi f}{C} L_i \sin\left(\varphi - \varphi_i\right)\right) \end{aligned} \tag{2}$$

where f and φ represent the frequency and aspect angle, respectively. f_c is the center frequency, C is the propagation velocity of electromagnetic wave, and q denotes the number of scattering centers. Θ is the set of all parameters, while the parameter vector describing the ith scattering center is defined as $\theta_i = \{A_i, x_i, y_i, \alpha_i, L_i, \varphi_i, \gamma_i\}$, where each parameter carries specific physical and geometric semantics. Specifically, A_i represents the complex amplitude intensity of the scattering center; (x_i, y_i) determines the relative spatial position of the scattering point in the image domain; α_i is the frequency dependence factor, which reveals the geometric curvature attributes of the scattering source (e.g., $\alpha_i \approx 1$ typically corresponds to a flat plate or cylindrical surface, while $\alpha_i \approx 0.5$ corresponds to an edge diffraction structure); L_i and φ_i describe the effective length and main orientation angle of the distributed scatterer, respectively; γ_i characterizes the dependence of scattering intensity on the aspect angle.

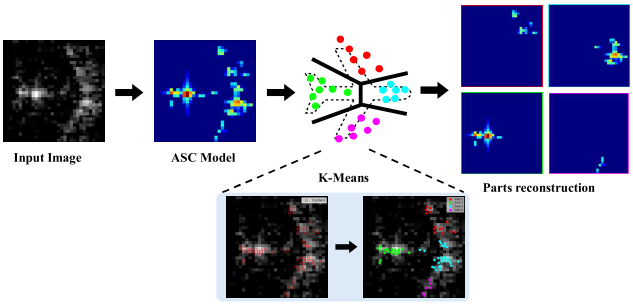


Fig. 2. Target component extraction leveraging the ASC model.

Based on the above model, parameter estimation is essentially transformed into a nonlinear optimization problem, i.e., inverting Θ by minimizing the difference between the observed signal and the model-reconstructed signal. Existing extraction methods cover two main approaches: image domain and frequency domain [45], [46]. We adopt a gradient-based optimization algorithm to balance extraction precision and computational efficiency [47]. During parameter estimation, the aspect dependence factor γ_i is directly set to zero due to its typically negligible value, thereby simplifying calculations and enhancing algorithm stability.

After obtaining the sparse parameter set describing the target, we employ the K-means clustering algorithm to perform unsupervised grouping on the discrete parameter set, as shown in Fig. 2. This method divides Θ into K nonoverlapping subsets $\{\Theta^{(1)}, \ldots, \Theta^{(K)}\}$ based on the similarity of spatial neighborhood distance and physical attributes. Here, $\Theta^{(1)}$ through $\Theta^{(K)}$ represent the total parameter sets for the 1st to the Kth groups, respectively. The value of K directly determines the total number of these target parts. Finally, by mapping each clustered parameter subset back to the image domain, a series of target component feature maps T_1 to T_K can be reconstructed. These feature maps represent collections of scattering centers with similar electromagnetic scattering characteristics, thereby achieving effective segmentation of target parts based on local physical properties.

It should be noted that the selection of the part quantity K in our method is primarily determined based on the prior physical topological structure of the target and the clustering distribution of ASCs. For the aircraft targets, setting K to 4 is a relatively appropriate choice to cover the relative topological structures of the nose, tail, left wing, and right wing.

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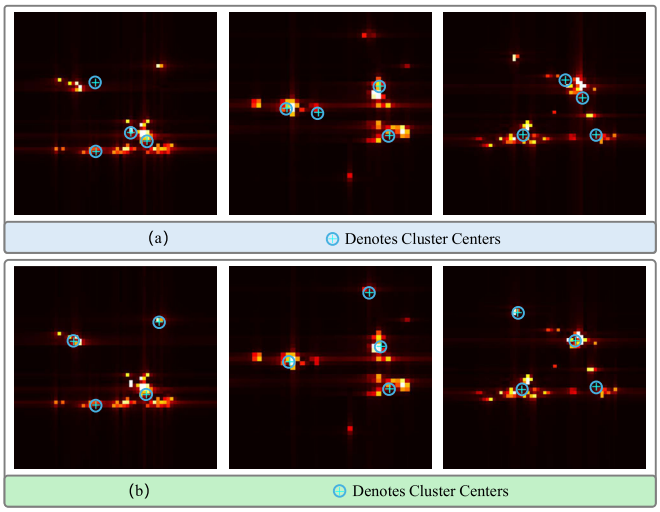


Fig. 3. Comparison of clustering performance: image-based versus multidimensional physical parameter integration. (a) $\{x, y, A\}$ cluster. (b) $\{x, y, \Phi, L, \alpha, A\}$ cluster.

Furthermore, traditional clustering strategies are often limited to the 3-D subspace of spatial coordinates (x, y) and amplitude A. This traditional process can be formulated as

$$S_{\text{trad}} = \text{KMeans}(x, y, A) \tag{3}$$

where S_{trad} denotes the generated cluster set. Such a partition, relying solely on spatial and intensity information, only captures geometric proximity within the image domain while potentially overlooking the intrinsic physical associations among scatterers. As shown in Fig. 3(a), the cluster centers of the three aircraft are located in close proximity. In contrast, incorporating the three key physical parameters of frequency dependence factor α, distribution length L, and orientation angle φ into the clustering process can be expressed as

$$S_{\text{phy}} = \text{KMeans}\,(x, y, A, \alpha, L, \varphi) \tag{4}$$

where S_{phy} denotes the generated cluster set. This essentially introduces the target's geometric topology type (such as edge, plane, or corner reflector) and distribution shape as strong constraints into the feature grouping process. This enhances the separability and discriminability of the cluster centers. As shown in Fig. 3(b), the cluster centers of the three aircraft are more widely separated than those in Fig. 3(a). This process, combined with the subsequent parametric reconstruction that maps the physical parameter set back to the image domain, collectively establishes the central role of physical prior information in the feature generation stage.

C. TSCM Module

1) Definition of Part Semantic Consistency: Existing research [38] has demonstrated that utilizing unsupervised clustering algorithms such as K-means can effectively aggregate discrete ASCs into target components with local characteristics. However, this clustering paradigm based on independent optimization of samples has a fundamental limitation: the discovered components lack a global canonical ordering.

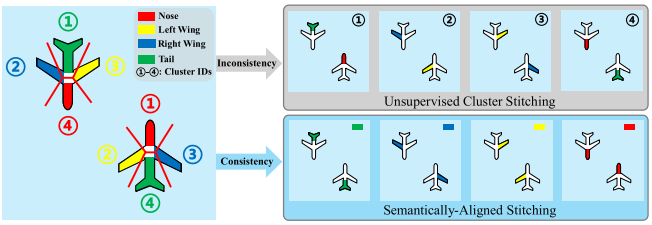


Fig. 4. Illustration of part semantic consistency.

Specifically, for the kth cluster in the ith sample obtained after K-means clustering, in the absence of semantic consistency constraints, the index k serves merely as a random nominal identifier, rather than a semantic descriptor with fixed physical meaning. We define this phenomenon as the label permutation problem. For example, the cluster with index $k = 1$ may correspond to the tail region in sample A, while in sample B, it may correspond to the wing due to initialization randomness or target pose variations.

This cross-sample semantic inconsistency poses severe challenges to downstream supervised learning, as shown in Fig. 4. If these raw clustering results are directly used for training, the neural network will receive conflicting supervisory signals targeting different physical structures on the same feature channel. This semantic entanglement prevents the network from learning part-specific features, causing them to collapse into fuzzy statistical averages of features of all components. Consequently, the feature representation degenerates into global target characteristics, leading to the failure of part-level physical injection.

To address the aforementioned problem, we formally define part semantic consistency as the ability to establish a stable, unique bijective mapping between feature channel index k and specific physical semantic categories s across the entire dataset.

Suppose we aggregate the kth target component feature map T_k of all N samples in the dataset to construct a cross-sample composite feature set S_k

$$S_k = \text{Stitch}\left(\left\{T_k^{(1)}, T_k^{(2)}, \ldots, T_k^{(N)}\right\}\right) \tag{5}$$

where $T_k^{(i)}$ (for $i = 1, 2, \ldots, N$) denotes the reconstructed part-level feature map corresponding to the kth semantic cluster of the ith sample. The $Stitch(\cdot)$ operator serves to combine individual component feature maps from N samples into a unified representation for dataset-wide analysis.

Under this definition, inconsistency implies that S_k contains a semantic mixture of different parts. For instance, the set mixes tail features from some samples with wing features from other samples, leading to confusion in the semantic space. Consistency requires that S_k can only contain the semantics of a single part; for example, all target component feature maps represent the tail.

This semantic consistency is the prerequisite for realizing interpretable physical injection. It forces the feature representation space of the network to be decoupled based on objective physical topological structures rather than arbitrary mathematical indices, thereby enabling the network to learn

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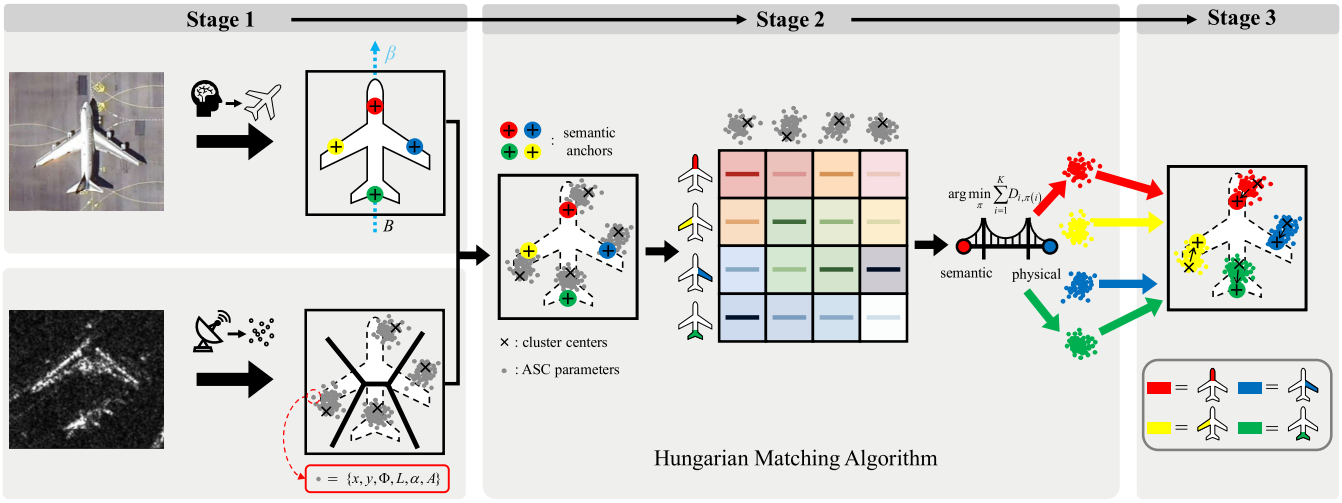


Fig. 5. Topology-aware semantic matching strategy.

robust, distinguishable part-level discriminative features with clear physical meanings.

2) Topology-Aware Semantic Matching Strategy: To ensure part semantic consistency, we propose a novel matching algorithm that aligns unsupervised physical clustering with topologically determined semantic keypoints. As shown in Fig. 5, this process comprises the following key steps.

1) *Configuration of Semantic Anchors:* Leveraging the oriented bounding box B and pose angle β provided by the annotations, we define a local coordinate system and calculate the ideal pixel coordinates for different semantic parts. In this study, the number of semantic parts is set to $K = 4$, specifically comprising the nose, tail, left wing, and right wing. The nose keypoint is located at 60% of the major axis vector, while the tail keypoint is at 10%, and the left and right wings are located on the two sides. This reflects the typical geometric shape of aircraft. These coordinates form an ordered set of semantic anchors $P_{\text{sem}} = \{p_1, p_2, p_3, p_4\}$, where p_1, p_2, p_3, and p_4 correspond to the four semantic parts: the head, tail, left wing, and right wing, respectively. These represent normalized positions derived from prior geometric knowledge.
2) *Clustering Based on Physical Parameters:* Simultaneously, we extract ASCs from complex SAR data and then apply K-means clustering in the parameter space $\vartheta = \{x, y, A, L, \alpha, \varphi\}$. This process generates a set of unordered physical parameters clusters $\Theta_{\text{phys}} = \{\Theta^{(1)}, \ldots, \Theta^{(K)}\}$ and corresponding physical cluster centers $C_{\text{phys}} = \{c_1, c_2, \ldots, c_K\}$, representing the main scattering mechanism regions.
3) *Global Optimal Matching Based on Hungarian Algorithm:* To resolve the correspondence between the unordered physical parameters clusters Θ_{phys} and the ordered set of semantic anchors P_{sem}, we construct a cost matrix $D \in \mathbb{R}^{K\times K}$, where each element D_{ij} represents the Euclidean distance between the ith semantic anchor p_i and the jth cluster center c_j (where $1 \le i, j \le K$)

$$D_{ij} = \| p_i - c_j \|_2 . \tag{6}$$

Then, we employ the Hungarian algorithm to solve for the optimal permutation π^* that minimizes the total spatial difference

$$\pi^* = \arg\min_{\pi} \sum_{i=1}^{K} D_{i,\pi(i)} \tag{7}$$

where π denotes a possible permutation mapping the physical cluster centers to the semantic anchors, and π^* represents the optimal assignment that minimizes the total spatial distance. This semantic matching strategy realizes the automated alignment of physical features and semantic concepts, establishing a mapping bridge between physical scattering features and high-level semantic concepts.

4) *ASC Model-Based Part Structure Mask Construction:* The objective is to extract part-level supervisory information with distinct physical semantics from complex SAR scattering signals and transform the continuous scattering intensity distribution into explicit binary geometric constraints. To this end, a semantic mask generation strategy based on the ASC model is proposed. While original SAR images provide rich scattering intensity information, directly utilizing continuous intensity maps as supervisory signals often forces the network to overfit to specific amplitude values, thereby neglecting the underlying spatial geometric morphology of the targets. Therefore, we propose transforming these continuous amplitude representations into discrete binary masks, which serve as robust geometric priors for the network. The detailed reconstruction process is provided in Algorithm 1.

The scattering center parameters of the target are first clustered and semantically matched using the aforementioned matching algorithm. Subsequently, according to the optimal permutation, the parameter set $\Theta^{(k)}$ of the kth specific semantic cluster (such as a wing component) is inversely mapped back to the image domain to reconstruct the target component feature map T_k for each independent part. Finally, T_k is stitched and binarized to generate the final semantically

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Algorithm 1 Topology-Guided Part-Consistent Mask Generation

Input: Complex SAR image I_{com}, annotations Y, part quantity $K = 4$.
Output: Part mask ground truth M_{gt}.
Initialize four empty global canvases M_k for $k = 1$ to K.
for each target object t in annotations Y **do**
// **Step 1: Data Preparation**
Crop target patch E_{patch} using bounding box B_t.
// **Step 2: Clustering Based on Physical Parameters**
Extract ASC parameters $\vartheta = \{x, y, A, L, \alpha, \varphi\}$ from E_{patch}.
Perform K-Means to get physical cluster centers C_{phys}.
// **Step 3: Configuration of Semantic Anchors**
Parse attitude angle theta and dimensions from B_t.
Calculate ideal semantic anchors P_{sem} using topology model.
// **Step 4: Global Optimal Matching**
Compute cost matrix D, where $D_{ij} = \| p_i - c_j \|_2$, $1 \le i, j \le K$
Solve for the optimal assignment using Hungarian Algorithm:
$\pi^* = \arg\min_\pi \sum_{i=1}^{K} D_{i,\pi(i)}$.
// **Step 5: Reconstruction & Stitching**
for $k = 1$ to K **do**
Retrieve $\Theta^{(k)}$ belonging to optimal assignment π^*.
Reconstruct $\Theta^{(k)}$ to image domain to obtain target component feature map T_k.
Stitch T_k onto global canvas M_k at original coordinates.
end for
end for
Binarize all canvases to generate part mask ground truth M_{gt}.
return M_{gt}

consistent part mask ground truth M_{gt}, which is used to guide the reconstruction network. This mask thoroughly strips away amplitude information susceptible to interference, and its nonzero regions solely characterize the geometric contours and positions of specific physical parts. This processing compels the network to focus on learning the spatial distribution and topological connection relationships of various parts during the training process, rather than mere texture details.

D. Physics-Guided Reconstruction and Injection Network

As the core execution unit of the PRIF framework, this article proposes the PRI-Net. To fully leverage physical supervisory information to enhance detection performance, the network is constructed based on a multitask learning [43] paradigm to form a collaborative architecture. It consists of two core tasks: the reconstruction task, responsible for recovering physical structures, and the detection task, responsible for target localization and recognition. These two task branches constitute a collaborative network guided by deep physical priors and data perception through a physical information injection mechanism.

1) Task A: PPRB: In terms of reconstruction supervision, our framework facilitates a transition from holistic feature recovery to part-level semantic decoupling. Existing general CV methods or most SAR detection reconstruction works often treat SAR data as noisy optical images, and their reconstruction goals are mostly to recover dense and continuous spatial pixel intensities or to predict the holistic macroscopic contours of targets. This signal-level or pixel-level reconstruction remains merely at the visual appearance, ignoring that SAR images are essentially 2-D projections of electromagnetic scattering mechanisms.

Furthermore, holistic masks are easily corrupted by clutter. Our framework upgrades traditional image recovery to the decoupling and internalization of electromagnetic physical semantics, introducing multidimensional physical parameters to decouple the target into part-level masks representing different physical mechanisms. Through the semantic consistency matching module, we ensure that the reconstruction supervision provides a topologically ordered structural penalty, significantly mitigating the ambiguity of part semantic permutation across samples. This fine-grained decoupling provides a robust local feature enhancement mechanism amid massive background noise, playing a global filtering role that traditional holistic mask methods cannot achieve.

On one hand, to internalize physical priors into feature representations, we design a progressive feature reconstruction path to deconstruct fine-grained physical structural contexts from deep features. As shown in the bottom-right corner of Fig. 1, adopting a top–down recursive strategy. The physical feature representation is first initialized at the highest pyramid level as follows:

$$R_5 = \text{Conv}_{\text{lat}}(P_5). \tag{8}$$

Subsequently, the refined physical feature R_i is recursively generated via the attention refinement module (ARM)

$$R_i = \text{Conv}_{\text{lat}}(P_i) \odot \left(1 + \sigma\left(\text{Conv}_{\text{attn}}\left(\text{Upsample}\left(R_{i+1}\right)\right)\right)\right) \quad i \in \{4, 3, 2\} \tag{9}$$

where

$$\text{Conv}_{\text{lat}}(\cdot) = \text{Dropout}\left(\text{ReLU}\left(\text{GN}\left(\text{Conv}_{3\times3}(\cdot)\right)\right)\right) \tag{10}$$

$$\text{Conv}_{\text{attn}}(\cdot) = \text{GN}\left(\text{Conv}_{1\times1}(\cdot)\right). \tag{11}$$

Upsample$(\cdot)$ denotes the bilinear interpolation upsampling, σ is the Sigmoid function, and $\odot$ represents the elementwise Hadamard product. P_i and R_i represent the ith FPN layer and the recursively generated physically refined feature of the ith layer, respectively. As shown in the formula, Conv_{lat} is a lateral feature extraction sequence containing 3×3 convolution, group normalization (GN), ReLU activation, and spatial Dropout, while $\text{Conv}_{\text{attn}}$ is a 1×1 convolution preprocessing unit used to generate attention weights. This formulation utilizes the residual attention mechanism within the ARM module, enabling deep semantics to dynamically modulate shallow details, thereby gradually restoring the physical geometric topology within the feature space. Subsequently, the final refined feature R_2 is fed into the reconstruction head to output the refined reconstruction mask M_r

$$M_r = \sigma\left(\text{Head}_{\text{recon}}(R_2)\right) \tag{12}$$

$$\text{Head}_{\text{recon}}(\cdot) = \text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Conv}_{3\times3}(\cdot)\right)\right). \tag{13}$$

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It is worth noting that the reconstruction of M_r is solely used to internalize electromagnetic scattering information within the network to assist the detection task. It is not used as a physical feature map directly injected into the detection branch.

On the other hand, to provide explicit physical prior guidance to the detection task, the part-aware probability mask M^i_{part} of the ith layer are directly decoded via the guide head

$$M^i_{\text{part}} = \sigma\left(\text{Head}_{\text{guide}}\left(P_i\right)\right). \tag{14}$$

Specifically, M^i_{part} signifies the part-aware probability mask generated at scale level i. The $\text{Head}_{\text{guide}}(\cdot)$ serves to refine the feature representation. Specifically, it is implemented by a single 3×3 convolutional layer. This part-aware probability mask serves as the key to multitask collaboration. It not only compels the network to learn physical topological information under the strong supervision of the reconstruction loss but also serves as an explicit structured prior injected into the detection branch to generate physically enhanced features with strong discriminability.

2) Task B: PDDB: In the dimension of physics-guided feature fusion, we accomplished the transition from rigid feature concatenation and stacking to explicit decoupled semantic attention. Most existing fusion methods rely on implicit global fusion, such as directly and roughly concatenating ASC parameters, or fusing intermediate-layer global spatial attention with ambiguous meaning, which easily leads to semantic aliasing between different physical components, such as wings and nose, and background clutter in the same feature channels.

In contrast, our part attention injection module (PAIM) extracts the reconstruction layer with the most intuitive and highest-level physical structural semantics for fusion. By integrating the mathematical constraints of grouped convolution, we transform this layer into a spatial attention map, forcing the establishment of a mapping relationship between specific physical part masks and specific feature channel subsets. We are not doing simple feature addition but utilizing precise physical topology to recalibrate the detection stream, explicitly isolating the semantic expressions of different physical components and preventing feature entanglement from the foundational tensor level.

To effectively fuse physical priors, this branch adopts a dual-drive architecture comprising a data-driven feature extraction module (FEM) and a physics-driven PAIM. This branch generally follows the dual-drive interaction concept [19], but differs fundamentally in that the physics drive in our physics-driven stream does not originate from predefined static priors, but directly from the dynamic reconstruction output of Task A.

Formally, let P_i be the input feature of the branch. This architecture achieves complementary enhancement through parallel dual-stream processing. Specifically, the FEM aims to extract global features. It adopts the squeeze-and-excitation (SE) mechanism [48] to calibrate feature responses through explicit channel modeling. The calculation process for the data-driven feature F_{data} and its internal components is defined as follows:

$$F_{\text{data}} = F_{\text{SE}}\left(\text{Conv}_{\text{data}}\left(P_i\right)\right) \tag{15}$$

$$\text{Conv}_{\text{data}}(\cdot) = \left(\text{ReLU}\left(\text{Conv}_{3\times 3}(\cdot)\right)\right) \tag{16}$$

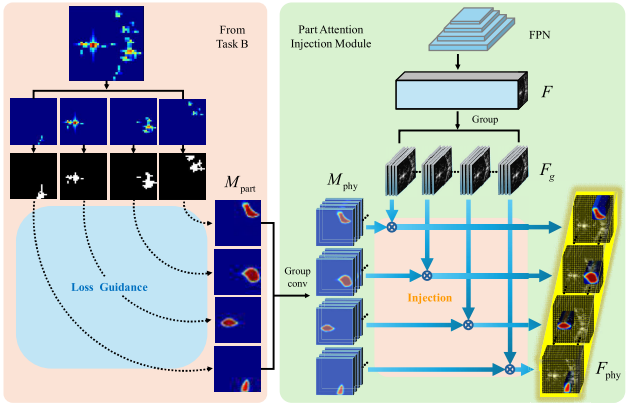


Fig. 6. PAIM.

$$F_{SE}(x) = x \cdot \sigma\left(\text{Conv}_{1\times 1}\left(ReLU\left(\text{Conv}_{1\times 1}\left(\text{GAP}(x)\right)\right)\right)\right) \tag{17}$$

where σ represents the Sigmoid activation function, and GAP denotes the global average pooling.

As shown in the formula, $\text{Conv}_{\text{data}}$ first performs spatial feature preprocessing via 3×3 convolution, followed by F_{SE} utilizing the channel attention mechanism to adaptively recalibrate feature weights, thereby extracting global features rich in context information.

Complementarily, PAIM aims to extract and enhance local features. This module utilizes the dynamic part-aware probability maps M_{part} reconstructed by Task A as the attention source with physical semantics, as shown in Fig. 6. To address the dimension mismatch problem between the physical mask (K channels) and the feature map (N_c channels) while ensuring that physical priors maintain a high degree of semantic independence during the injection process, we propose a decoupled spatial attention mechanism. This mechanism divides the high-dimensional feature map into distinct subgroups, allowing each part mask to independently modulate its corresponding feature channels.

Benefiting from the strong supervision of the consistent semantic mask in Task A, the part-aware probability mask M_{part} maintains a high degree of semantic consistency. This provides a reliable physical basis for subsequent feature decoupling; otherwise, aliased masks would directly undermine the physical significance of the grouping operation.

Specifically, Task A generates M_{part}, which must be mapped into a high-dimensional feature space. However, conventional fully connected convolutions often lead to uncontrollable semantic mixing across different part channels. To address this, we employ grouped convolution as a projection operator to preserve physical semantics by strictly setting the number of groups to equal the number of parts. This design forces the establishment of a one-to-many physical mapping relationship. That is, each input physical part channel is only expanded and linearly mapped to a specific set of subchannels in the output feature map. The generated tensor is the decoupled physical attention map, denoted as M_{phy}, and its calculation formula is defined as

$$M_{\text{phy}} = \text{GConv}_{\text{expand}}\left(M_{\text{part}}\right) \tag{18}$$

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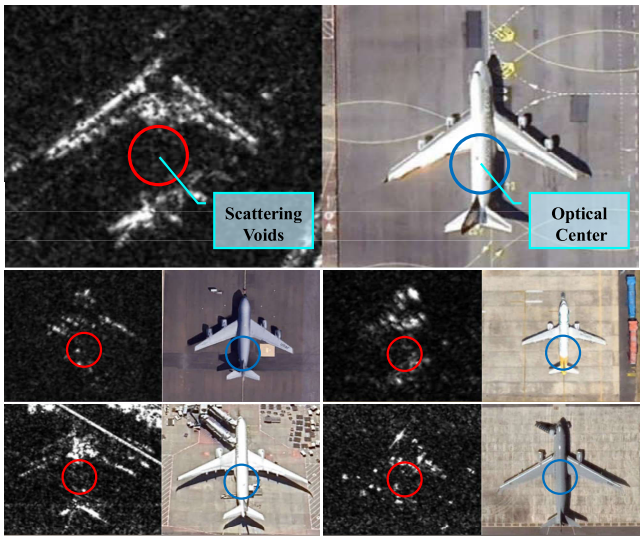


Fig. 7. Scattering characteristics of SAR aircraft targets: discrete scattering and scattering voids.

where $M_{\text{phy}} \in \mathbb{R}^{N_c \times H \times W}$, and different channel groups strictly correspond to and are controlled by different physical structures. Simultaneously, the input feature F is divided equally into $K = 4$ groups along the channel dimension to obtain grouped features F_g. Finally, we execute explicit spatial injection, i.e., using the decoupled physical attention map M_{phy} to perform physics-guided spatial modulation on the grouped features F_g

$$F_{\text{phy}} = F_g \odot M_{\text{phy}} \tag{19}$$

where F_{phy} denotes the physically modulated feature map generated after applying the explicit spatial injection. Through this operation, the feature map is explicitly physically reshaped in its spatial distribution. In nontarget regions, feature responses are suppressed by the zero values of the physical mask; whereas within specific channel groups, features are highlighted and activated only in the corresponding physical part regions, such as wings or nose. Finally, by fusing the dual-stream outputs, we obtain the physically enhanced detection feature $F_{\text{out}} = F_{\text{data}} + F_{\text{phy}}$. Subsequently, F_{out} is fed into the parallel FCOS classification and regression heads to complete the final detection prediction.

E. PALA Strategy and Hybrid Loss Function

1) PALA Strategy: Traditional anchor-free detection frameworks, such as FCOS, typically assign positive samples based on a geometric center prior, implicitly assuming that the geometric center of the target corresponds to a feature-rich region [13]. However, SAR aircraft targets exhibit significant scattering discreteness, with strong scattering sources primarily distributed along specific parts such as the nose and leading edges of wings. As shown in Fig. 7, these scattering sources present a noncontinuous and nonuniform spatial morphology. This spatial misalignment between the geometric center and the physical feature center often leads traditional strategies to assign positive samples to nonsignificant regions with weak echoes. Consequently, the network struggles to extract effective discriminative features, thereby triggering severe sample quality degradation.

To this end, this article designs a PALA strategy adapted to discrete scattering characteristics. This method aims to correct the sampling criterion from simple geometric location to physical entity alignment, effectively resolving the scattering voids problem. Specifically, we introduce the part mask ground truth M_{gt} generated by TSCM as a topological hard constraint during the assignment phase. A feature anchor point $z(x, y)$ is determined to be a positive sample Ω_{pos} if and only if it satisfies the dual conditions of geometric location and physical attribute

$$z(x, y) \in \Omega_{\text{pos}} \Leftrightarrow (z(x, y) \in \mathcal{B}) \wedge \left(z(x, y) \in M_{\text{gt}}\right) \tag{20}$$

where $\mathcal{B}$ denotes the spatial region defined by the oriented bounding box of the target.

To avoid the noise introduced by forced explicit prediction in the feature-sparse geometric center region, we discarded the explicit centerness branch, but retained the geometric centerness ground truth C_{z^*} to weight the regression loss. This implicit stability guidance mechanism prompts the network to prioritize the optimization of high-quality anchors within the PALA positive sample set that possess both clear physical features and geometric center stability. This strategy effectively balances physical saliency and regression stability, avoiding interference from the scattering voids while significantly improving the localization robustness of the detector.

2) Hybrid Loss Function: To synergistically optimize the physical reconstruction branch (Task A) and the physical injection detection branch (Task B), PRI-Net adopts a joint training strategy. The total loss $\mathcal{L}_{\text{total}}$ combines the detection loss $\mathcal{L}_{\text{det}}$ and reconstruction loss $\mathcal{L}_{\text{rec}}$ through a weighted summation

$$\mathcal{L}_{\text{ptotal}} = \mathcal{L}_{\text{det}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} \tag{21}$$

where the balancing coefficient λ_{rec} is empirically set to 0.4 to prevent the auxiliary task from dominating the gradient direction in the later stages of training, thereby ensuring the convergence stability of the main detection task.

For the detection branch, the total loss $\mathcal{L}_{\text{det}}$ is formulated as the sum of the classification loss $\mathcal{L}_{\text{cls}}$ and the rotation regression loss $\mathcal{L}_{\text{reg}}$

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}}. \tag{22}$$

The classification loss $\mathcal{L}_{\text{cls}}$ employs focal loss $\mathcal{L}_{\text{FL}}$ [12], which ensures that the classification branch is activated exclusively in part regions characterized by distinct scattering features. The formula is expressed as follows:

$$\mathcal{L}_{\text{cls}} = \frac{1}{N_{\text{pos}}} \sum_{z \in \Omega_{\text{all}}} \mathcal{L}_{\text{FL}}\left(p_z, c_z^*\right). \tag{23}$$

In the above equation, N_{pos} represents the total number of physical positive samples, used to normalize the loss; Ω_{all} represents the set of all feature anchor points; p_z and c_z^* represent the predicted classification confidence and the ground-truth class label at point z, respectively.

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The rotation regression loss $\mathcal{L}_{\text{reg}}$ utilizes the rotated intersection over union (IoU) loss, calculated solely for the physical positive sample set Ω_{pos}

$$\mathcal{L}_{\text{reg}} = \frac{1}{N_{\text{pos}}} \sum_{z \in \Omega_{\text{pos}}} C_{z^*} \cdot \mathcal{L}_{\text{RIoU}} \left(\mathcal{B}_z, \mathcal{B}_{z^*}\right) \tag{24}$$

where $\mathcal{B}_z$ and $\mathcal{B}_{z^*}$ denote the predicted 5-D rotated bounding box vector and its corresponding ground truth at point z, respectively; $\mathcal{L}_{\text{RIoU}} = 1 - \text{IoU}_{\text{rot}}(\mathcal{B}_z, \mathcal{B}_{z^*})$ calculates the loss of the rotated IoU between the two, where $\text{IoU}_{\text{rot}}(\cdot)$ denotes the function that computes the exact IoU between two oriented bounding boxes [15]. C_{z^*} is the ground-truth geometric centerness [13], serving here as a pixelwise regression weight.

For the reconstruction branch, to address the challenges of the extremely small pixel proportion (<1%) and blurred edges of SAR targets, we constructed a composite constraint mechanism comprising a multiscale feature alignment loss $\mathcal{L}_{\text{align}}$ and a high-fidelity structural integrity loss $\mathcal{L}_{\text{struct}}$. First, the multiscale feature alignment loss $\mathcal{L}_{\text{align}}$ is defined as the sum of the weighted mean squared error between the prediction maps at each FPN level and the ground-truth masks

$$\mathcal{L}_{\text{align}} = \sum_{i=2}^{5} W \odot \left(M_{\text{part}}^{i} - M_{\text{gt}}\right)^2 \tag{25}$$

where i represents the level index of the feature pyramid and W is a preset balancing weight matrix (foreground weight is set to 10.0, background weight is set to 1.0). M_{part}^{i} is bilinearly upsampled to the same size as M_{gt} for consistency during training.

Second, the high-fidelity structural integrity loss $\mathcal{L}_{\text{struct}}$ targets the final output high-resolution refined reconstruction mask M_r, employing a linear combination of dice loss $\mathcal{L}_{\text{Dice}}$ [41] and a pixelwise focal loss $\mathcal{L}_f$ to simultaneously enhance global topological consistency and local edge details

$$\mathcal{L}_{\text{struct}} = \mathcal{L}_{\text{Dice}} \left(M_r, M_{\text{gt}}\right) + \lambda_f \mathcal{L}_f \left(M_r, M_{\text{gt}}\right) \tag{26}$$

where $\mathcal{L}_f$ shares the same mathematical core as $\mathcal{L}_{\text{FL}}$ but operates in an elementwise manner over the 2-D spatial mask. M_r is bilinearly upsampled to the same size as M_{gt} for consistency during training.

We set the adjustment factor λ_f to 20.0 to balance the gradient magnitudes of the two losses. Finally, the total loss of the reconstruction branch $\mathcal{L}_{\text{rec}}$ is

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{struct}}. \tag{27}$$

III. Experiments

This section conducts systematic experiments on the FAIR-CSAR and RSAR datasets to verify the effectiveness of the PRIF framework in complex SAR scenarios.

A. Datasets and Settings

1) FAIR-CSAR Dataset: This dataset is derived from the FAIR-CSAR benchmark dataset [52], comprising high-resolution complex-valued SAR imagery. We specifically utilized the spotlight mode subset and filtered the data to retain images containing only 12 types of aircraft targets, constructing an aircraft subdataset. The filtered dataset contains 2668 images with a total of 22 098 aircraft instances. The data split follows the original protocol, divided into a training–validation set (2289 images and 19 213 instances) and a test set (379 images and 2885 instances). During the training phase, 10% of the training–validation set is randomly selected as the validation set, and the model achieving the highest mean average precision (mAP) on the validation set is selected for testing.

2) RSAR Dataset: The RSAR dataset was proposed at the CVPR 2025 conference [53]. We constructed an aircraft subdataset by retaining images containing only the aircraft category. It consists of 883 images containing 7823 aircraft targets. The data split follows the original benchmark settings, including a training set (795 images and 6880 instances), a validation set (44 images and 449 instances), and a test set (44 images and 494 instances). Similarly, we selected the optimal model based on the best mAP performance on the validation set.

3) Experimental Settings: We train the model for 100 epochs with a batch size of 2, starting with an initial learning rate of 0.001, which decays to 10% at the 60th and 80th epochs, respectively. Model optimization is performed using stochastic gradient descent (SGD), configured with a momentum of 0.9 and a weight decay coefficient of 0.001. To ensure fair comparison, all methods are implemented based on the MMRotate toolkit [54].

B. Evaluation Metrics

To assess the efficacy of our proposed method in SAR detection, a multifaceted evaluation framework is established using precision (P), recall (R), the $F1$-score, and mAP. Specifically, P and R quantify the model's robustness against false positives (FPs) and missed detections, while the $F1$-score provides a balanced harmonic mean of these two metrics. These indicators are derived from true positive (TP), FP, and false negative (FN) counts based on the IoU criterion, as defined below

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = \frac{2 \times \text{P} \times \text{R}}{\text{P} + \text{R}}. \tag{28}$$

As the core criterion for measuring the comprehensive performance of multiclass detectors, mAP is defined as the arithmetic mean of the AP (area under the P–R curve) for all categories. Its calculation formula is

$$\text{mAP} = \frac{1}{|N_{\text{cls}}|} \sum_{i=1}^{|N_{\text{cls}}|} \text{AP}_i \tag{29}$$

where $|N_{\text{cls}}|$ denotes the overall category count. We selected two key metrics. The mAP at an IoU threshold of 0.50 is employed to evaluate the basic detection capability, while the average value calculated over the IoU interval from 0.50 to 0.95 is used to measure the overall robustness of the model under varying levels of localization strictness.

C. Comparison Results and Analysis

1) Comparison With State of the Art: To validate the effectiveness of the proposed PRIF framework in complex SAR

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TABLE I
COMPARISON WITH OTHER METHODS

Method	mAP50	mAP50:95	Precision	Recall	F1
Faster R-CNN [11]	0.411	0.172	0.498	0.496	0.497
RetinaNet [12]	0.400	0.169	0.525	0.432	0.474
RoI Transformer [49]	0.410	0.183	0.506	0.502	0.504
ATSS [50]	0.386	0.174	0.489	0.465	0.477
GWD [17]	0.413	0.169	0.536	0.456	0.493
R3Det [16]	0.425	0.181	0.528	0.510	0.519
ReDet [51]	0.326	0.143	0.427	0.416	0.422
S2ANet [15]	0.417	0.188	0.510	0.483	0.496
KLD [18]	0.378	0.140	0.517	0.447	0.479
UCR [53]	0.259	0.086	0.450	0.359	0.399
RQFormer [55]	0.399	0.183	0.530	0.459	0.492
RHINO [56]	0.445	0.191	0.518	0.493	0.505
Ours	**0.468**	**0.205**	**0.568**	**0.528**	**0.547**

TABLE II
GENERALIZATION VERIFICATION ON RSAR DATASETS

Method	mAP50	mAP50:95	Precision	Recall	F1
Faster R-CNN [11]	0.738	0.369	0.814	0.850	0.832
RetinaNet [12]	0.758	**0.432**	0.836	0.838	0.837
RoI Transformer [49]	0.753	0.426	0.832	0.842	0.837
ATSS [50]	0.762	0.382	0.822	0.824	0.823
GWD [17]	0.778	0.414	0.840	0.828	0.834
R3Det [16]	0.754	0.404	0.828	0.838	0.833
ReDet [51]	0.769	0.430	0.827	0.822	0.824
S2ANet [15]	0.770	0.409	0.839	0.842	0.840
KLD [18]	0.781	0.426	**0.843**	0.818	0.830
UCR [53]	0.716	0.330	0.806	0.798	0.802
RQFormer [55]	0.779	0.431	0.830	0.808	0.818
RHINO [56]	0.700	0.387	0.789	0.781	0.785
Ours	**0.790**	0.402	0.827	**0.858**	**0.842**

scenarios, we conducted a comprehensive comparison between this method and a series of mainstream and advanced oriented object detection algorithms, as shown in Table I. These comparative methods can be primarily categorized into one-stage, two-stage, and Transformer-based approaches. The one-stage methods include the anchor-based RetinaNet [12], S2Anet [15], and R3Det [16], as well as the anchor-free method ATSS [50]. In addition, we incorporate KLD [18], GWD [17], and the SAR-specific method UCR [53]. The two-stage methods, which predominantly rely on anchor-based mechanisms, encompass Faster R-CNN [11], RoI Transformer [49], and ReDet [51]. Regarding Transformer-based approaches, we introduce RQFormer [55] and RHINO [56], two DETR-based models.

As shown in Table I, experimental results indicate that the proposed method achieves the best performance across all metrics. Specifically, the core metric mAP50 reaches 0.468, significantly outperforming the second-best RHINO (0.445) and third-best R3Det (0.425), representing an improvement of 2.3% and 4.3%. This demonstrates that introducing ASC physical topology priors can effectively compensate for the deficiencies of relying solely on data-driven feature extraction, significantly enhancing the model's feature discrimination capability for SAR aircraft targets. Mechanistically, by transforming decoupled part-level physical semantics into explicit spatial attention, the model functions as a robust global spatial filter, sharply isolating true electromagnetic target signatures from severe background clutter.

The detection results in Fig. 8 verify the superiority of the proposed method. By transforming decoupled physical semantics into an explicit spatial attention prior, the global spatial filtering mechanism effectively suppresses severe background clutter, ensuring that both the missed detection rate and the false alarm rate remain at a notably lower level compared to other methods. Furthermore, the proposed method exhibits high detection rates and localization precision specifically for small targets. This is primarily attributed to the high-fidelity refined reconstructed feature maps. For discrete small targets with extremely weak scattering responses, these high-fidelity maps effectively preserve and recover their fine-grained physical structural details, successfully preventing their fragile part-level semantics from being diluted or lost during the deep network's downsampling and feature extraction processes.

Furthermore, although methods such as S2ANet and R3Det have achieved decent results through visual feature alignment, they lack an understanding of electromagnetic scattering mechanisms. In contrast, PRIF effectively bridges the gap between pure data-driven paradigms and the underlying electromagnetic scattering mechanisms by integrating the ASC model with deep learning.

2) Generalization Verification on Other Datasets: To verify the generalization capability of the proposed method, we extended the experiments to the RSAR dataset, which contains only amplitude data. To address the lack of phase information and pose angle annotations in this dataset, we performed an adaptive dimensionality reduction on the physics-guided strategy: the high-dimensional ASC model was degenerated into a point scattering center model based on amplitude peaks; simultaneously, the target pose was coarsely estimated by calculating the energy density distribution along the major axis of the bounding box (the side with higher energy was determined as the nose), thereby achieving a preliminary alignment of part semantics.

The experimental results, as presented in Table II, indicate that even under this degraded configuration with restricted data modalities and coarse-grained alignment, the proposed method still achieves the top ranking across three metrics. Specifically, the core metric mAP50 reaches 0.790 (outperforming the second-best method, KLD, by 0.9%), and the recall reaches as high as 0.858. This is not only attributed to the effectiveness of our physical reconstruction and injection mechanism, but also provides compelling evidence that ordered semantic constraints are superior to unordered feature states. Incorporating even coarse semantic topological constraints effectively bolsters the network's proficiency in representing discrete targets. Furthermore, this indicates that if full complex data and pose angle annotations become available in the future to achieve finer semantic alignment, the detection accuracy and physical interpretability of this method are expected to be further significantly enhanced.

D. Ablation Studies

To verify the effectiveness and synergistic mechanism of each component in the PRIF framework, this section designs a hierarchical ablation study from macro to micro levels. Considering the complex logic among the internal submodules

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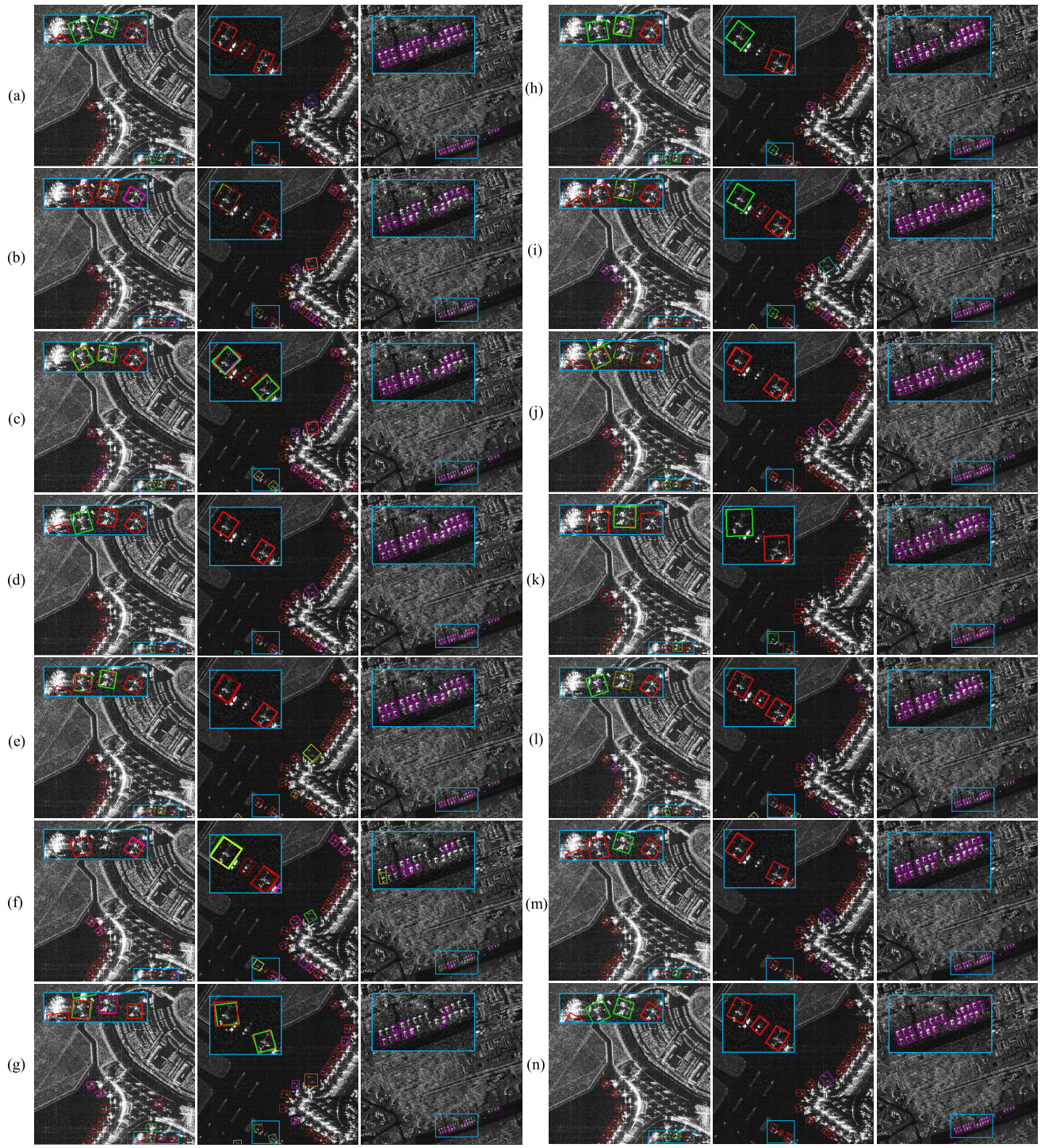


Fig. 8. Comparison of detection results with other methods. (a) Groundtruth. (b) RetinaNet. (c) Faster R-CNN. (d) RoI Transformer. (e) ATSS. (f) ReDet. (g) KLD. (h) GWD. (i) S2ANet. (j) R3Det. (k) UCR. (l) RQFormer. (m) RHINO. (n) Ours.

(PPRB, FEM, and PAIM) of PRI-Net, the experiment is divided into two parts. First, at the macro level, PRI-Net, TSCM, and PALA are taken as a whole for system-level ablation to verify the complementary relationship among feature extraction, semantic alignment, and spatial sampling. Second, it delves into the interior of PRI-Net to conduct structural ablation on the submodules to reveal the dual-stream fusion mechanism.

The macro experiments reveal the independent gain mechanism of each module. As shown in Table III, independently introducing PRI-Net increases mAP50 to 0.438, proving that the physical prior enhances the implicit structural perception of targets. Independently introducing PALA increases recall to 0.502, and mAP50 reaches 0.428, because it abandons traditional center sampling and aligns with physical entities to initially capture weak discrete targets. Introducing the TSCM

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TABLE III
ABLATION EXPERIMENT OF PRIF

PRI-Net	PALA	TSCM	mAP50	mAP50:95	P	R	F1
✗	✗	✗	0.415	0.186	0.523	0.495	0.509
✗	✓	✗	0.428	0.188	0.530	0.502	0.516
✓	✗	✗	0.438	0.190	0.528	0.495	0.511
✓	✓	✗	0.445	0.196	0.531	0.511	0.521
✓	✗	✓	0.452	0.194	0.530	0.488	0.508
✓	✓	✓	**0.468**	**0.205**	**0.568**	**0.528**	**0.547**

TABLE IV
ABLATION EXPERIMENT OF PRI-NET SUBMODULES

PPRB	PDDB		mAP50	mAP50:95	P	R	F1
	PAIM	FEM					
✗	✗	✗	0.428	0.188	0.530	0.502	0.516
✓	✗	✗	0.454	0.197	0.529	0.504	0.516
✓	✓	✗	0.465	0.203	0.518	**0.537**	0.527
✓	✗	✓	0.450	0.190	0.525	0.485	0.504
✓	✓	✓	**0.468**	**0.205**	**0.568**	0.528	**0.547**

module on the basis of PRI-Net increases mAP50 to 0.452 by alleviating feature aliasing through channelwise semantic decoupling.

It is worth noting that when only combining PRI-Net and TSCM, positive samples easily fall into echo-free regions due to the continuation of traditional center sampling, leading to feature mismatch, and recall instead drops to 0.488. Under full configuration, this deep coupling from feature perception and semantic decoupling to spatial mapping enables the joint use of the three modules to generate a powerful synergistic amplification effect. The precision and recall jump to 0.568 and 0.528, respectively, and mAP50 reaches an optimal 0.468.

The micro ablation reveals the evolution process of the dual-stream architecture of PRI-Net. As shown in Table IV, PPRB significantly increases mAP50 from 0.428 to 0.454, proving that the physical reconstruction task strengthens the perception of the network regarding implicit physical structures. Subsequently, if only the data-driven module FEM is introduced, mAP50 drops slightly to 0.450, indicating that global context features dilute the sensitivity to local physical features to some extent. To resolve this conflict, the PAIM module is further introduced for explicit anchoring of local parts, making the global context of FEM and the local prior of PAIM complement each other. This successfully suppresses feature degradation and achieves synergistic enhancement, ultimately raising mAP50 to a peak of 0.468.

E. Verification of the Portability of Main Modules

To evaluate the universality of the physics-guided reconstruction and injection mechanism (PPRB, PDDB, and TSCM), we integrated these core components into two representative detection frameworks: RetinaNet and S2ANet. Table V tabulates the quantitative outcomes, demonstrating that these modules exhibit excellent plug-and-play characteristics and can significantly compensate for the deficiencies

TABLE V
VERIFICATION OF THE PORTABILITY OF MAIN MODULES

Method	mAP50	mAP50:95	P	R	F1
RetinaNet	0.400	0.169	0.525	0.432	0.474
RetinaNet+ours	**0.439**	**0.189**	**0.555**	**0.473**	**0.511**
S2ANet	0.417	0.188	0.510	**0.483**	0.496
S2ANet+ours	**0.437**	**0.195**	**0.548**	**0.483**	**0.514**

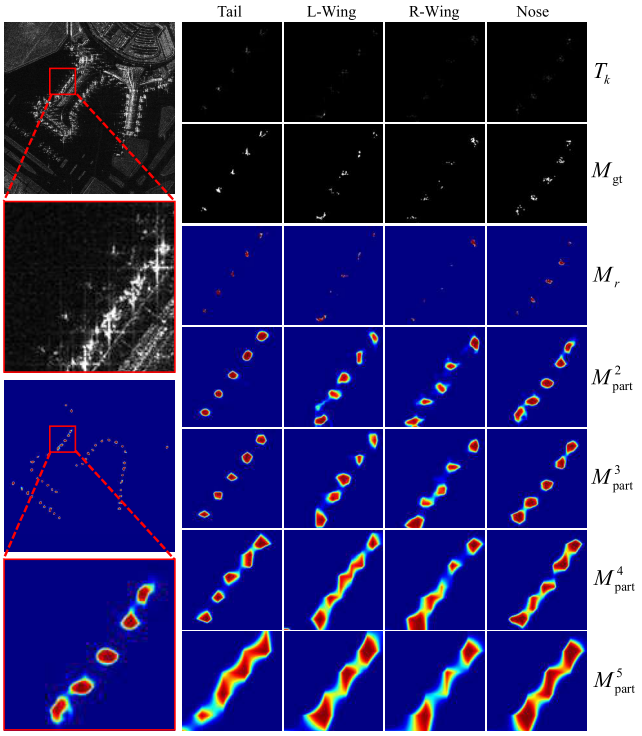


Fig. 9. Visualization of reconstruction results.

of purely data-driven networks: the mAP50 of RetinaNet improved by 3.9%, and that of S2ANet improved by 2.0%.

In-depth analysis of the metrics reveals that the performance gains manifest as comprehensive enhancements tailored to the baselines. For RetinaNet, both precision (+3.0%) and recall (+4.1%) increased significantly, yielding a 3.7% $F1$-Score boost. For S2ANet, precision surged by 3.8% while maintaining robust recall. This observation demonstrates that the decoupled spatial attention mechanism endows the network with a robust capability to perceive discrete target structures, effectively recovering weak targets missed by conventional feature extraction methods. Furthermore, these enhanced structure-aware features effectively suppress false alarms in background clutter, thereby significantly improving detection accuracy.

F. Visualization of Reconstruction Results

To verify the physical interpretability of the model, we visualized the key outputs of the reconstruction branch, as shown in Fig. 9. The top row, showing T_k and M_{gt}, demonstrates that, benefiting from the TSCM module, various parts (nose, wings, and tail) are effectively decoupled. The middle row, M_r, generated by progressive feature reconstruction, deconstructs

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TABLE VI
K VALUE ABLATION EXPERIMENT

K	mAP50	mAP50:95	P	R	F1
1	0.449	0.176	0.530	0.508	0.518
2	0.450	0.179	0.507	0.517	0.512
4	**0.468**	**0.205**	**0.568**	**0.528**	**0.547**
8	0.460	0.187	0.551	0.481	0.514

TABLE VII
SEMANTIC TOPOLOGY ABLATION EXPERIMENT

Semantic topology	mAP50	mAP50:95	P	R	F1
 Rectangular shape	0.455	0.201	0.504	0.517	0.51
 Vertical line shape	0.460	**0.205**	0.503	0.532	0.517
 T-shape	0.467	0.202	0.535	**0.536**	0.536
 Aircraft shape	**0.468**	**0.205**	**0.568**	0.528	**0.547**

fine-grained physical structural context. The bottom rows, from M^2_{part} to M^5_{part}, reveal the evolution of physical features across FPN levels. This method follows the scale assignment principle of FCOS, where physical features are explicitly injected only at levels matching the target scale.

As shown in the figure, for such small targets, shallow features retain rich high-frequency details and clear contours, serving as the dominant layer for physical injection; whereas deep features are blurred due to excessively large receptive fields and serve only as auxiliary support. Conversely, for large targets, deep features will present clear part topologies and play a dominant role. This strict adaptive alignment between physical injection and feature scale ensures that physical priors take effect at the optimal level, thereby significantly improving detection precision and recall.

G. Discussion

1) Comparison of Different K Values and Semantic Anchors: To verify the robustness of parameter settings, this section explores the selection mechanism of the part quantity K and semantic topology through ablation experiments. Following the divisibility requirement of grouped convolution dimensions and the convention that mainstream network channel numbers are mostly powers of 2, K is strictly limited to integer powers of 2 to ensure the validity of underlying tensor operations. Meanwhile, this article compares four topological layouts with different fitting degrees to verify the scientificity of the setting.

In the K value ablation experiment, the data reflects obvious undersegmentation and oversegmentation effects. As shown in Table VI, when K is 1 or 2, the extremely small value leads to physical semantic aliasing, making it impossible to finely decouple local features, which causes mAP50 and $F1$-scores to be at a low level. When K increases to 8, the over-segmentation phenomenon destroys the physical integrity of the target and introduces a large amount of fragment noise, leading to a fallback in mAP50 and a sudden drop in recall. The experiment proves that when $K = 4$, an ideal balance between architectural constraints and feature decoupling is achieved, obtaining an mAP50 of 0.468 and an $F1$-score of 0.547.

In the semantic topology ablation experiment fixing $K = 4$, four layouts from abstract to concrete are compared. Table VII shows that even if the rectangular topology deviates from the real shape, its mAP50 still reaches 0.455, revealing the underlying constraint mechanism: as long as a unified spatial mapping rule is followed, the orderliness of parts across samples can be forcibly established, breaking the randomness of unsupervised clustering. As the similarity between the topological structure and the target physical shape increases, the detection performance shows a positively correlated growth, with mAP50 steadily rising from 0.455 to 0.467. When utilizing an airplane shape topology that completely fits the physical distribution, the part semantic alignment reaches its extreme, maximizing intrachannel consistency and enhancing discriminative features among parts, ultimately driving precision to jump to 0.568.

2) Model Complexity Analysis: This section conducts a comprehensive quantitative verification of the computational overhead of the model. It should be particularly emphasized that the steps of ASC modeling, K-means clustering, and Hungarian matching are completely decoupled from the forward inference of the core network, serving solely as an offline process to generate mask supervision signals. This neither occupies training time nor increases the number of inference parameters and computational complexity. Only the operations of mask reconstruction and physical feature injection affect training, which mainly involve the spatial inverse mapping of parameters, and their complexity has a strict linear relationship with the resolution of the generated feature maps.

The computational complexity of the ASC modeling can be expressed as $O(K_{\text{asc}} \cdot (N_p log N_p + I \cdot N_p \cdot d_{\text{phy}}))$, where K_{asc} represents the total number of scattering centers to be extracted, N_p is the total number of points in the complex echo data, I represents the number of iterations in the parameter optimization process of a single scattering center, and d_{phy} is the dimensionality of the physical parameters to be solved. The complexity of the K-means clustering algorithm is $O(N_{\text{iter}} \cdot K \cdot K_{\text{asc}} \cdot d_{\text{phy}})$, where N_{iter} is the number of convergence iterations of the clustering algorithm, and K is the part quantity. The time complexity of the Hungarian matching algorithm is $O(K^3)$, where K also represents the number of parts. Regarding the time consumption distribution of data preprocessing operations, K-means clustering and Hungarian matching take extremely short time, and the time occupation is mainly concentrated on ASC modeling, where single target extraction takes approximately 14.68 s.

Table VIII shows that under the same P3–P7 detection levels, after introducing the physics-guided feature network, the model parameter volume only increases from 32.13M to 36.86M, FLOPs slightly increase by about 14G, and the

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TABLE VIII
MODEL COMPLEXITY ANALYSIS WITHOUT OFFLINE STEPS

Method	Detection level	mAP50	Params (M)	FLOPs (G)	FPS
Baseline	P3-P7	0.415	32.13	206.85	23.36
Baseline	P2-P5	0.411	31.61	560.04	15.65
ours	P3-P7	0.423	36.86	220.98	22.98
ours	P2-P5	0.468	32.21	612.19	13.97

inference frame rate only drops from 23.36 to 22.98. This set of comparisons confirms that the progressive reconstruction and grouped injection modules possess extremely high efficiency, and the physical stream fusion does not constitute an inference bottleneck.

Further analysis reveals that the main reason for the increased computational load lies in the selection of high-resolution levels. Targeting the discrete and tiny characteristics of SAR targets, the P2–P5 levels are crucial for capturing fine topologies. The experiment found that after the baseline model switches to high-resolution levels, FLOPs surge to 560.04G and the frame rate drops to 15.65, but mAP50 instead degrades to 0.411. This indicates that traditional networks are susceptible to background clutter interference when introducing shallow features and lack an effective filtering mechanism.

In contrast, the final version of this article achieves an optimal precision of 0.468 mAP50 with a low parameter volume of 32.21M under the P2–P5 levels. The comparison with the high-resolution baseline confirms that the proposed method can effectively suppress noise and transform high-resolution features into highly discriminative physical representations. Although processing P2 features fixes the inference frame rate at 13.97, this is not physical module redundancy but a reasonable computing power tradeoff to improve the perception precision of tiny targets.

3) Selection of Loss Function Hyperparameters: Aiming at the tuning problem of the multitask loss function, this section designs a control variable experiment. This experiment aims to balance the gradient contributions between the reconstruction auxiliary task and the main detection task, and to address the severe foreground and background imbalance problem in SAR images. The first phase fixes the foreground weight $W = 10$ to conduct a 2-D grid search to optimize the adjustment factors λ_{rec} and λ_f. The second phase conducts a 1-D ablation analysis on W under the optimal combination, aiming to quantitatively reveal the gradient game mechanism among various parameters.

Fig. 10 shows that in the 2-D experiment, fixing $W = 10$, the data present an evolution pattern of parameter coupling and an inverted U shape. When λ_{rec} is in the low value range of 0.1–0.2, the model fails to fully internalize the geometric topology due to the excessively weak reconstruction gradient, causing mAP50 to stagnate below 0.460. When it exceeds 0.6 or even reaches 1.0, the excessively large gradient of the auxiliary task interferes with the optimization of the detection task, degrading the performance to the range of 0.448–0.459. Regarding λ_f, when $\lambda_{rec} = 0.4$, if λ_f drops to 1, the model has insufficient penalty for tiny point details, and mAP50 drops to 0.456. If it rises to 80–100, the accuracy declines to 0.455

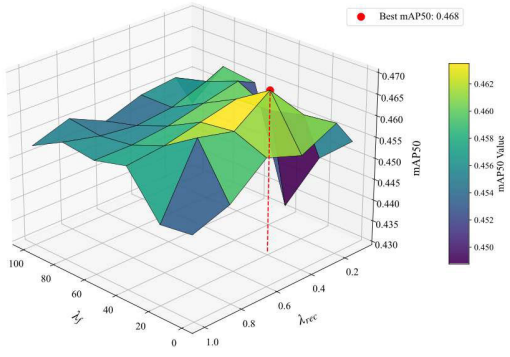


Fig. 10. Comparative experiment of λ_{rec} and λ_f.

TABLE IX
COMPARATIVE EXPERIMENT OF W

W	mAP50	mAP50:95	P	R	F1
1	0.456	0.196	0.512	0.525	0.519
10	**0.468**	**0.205**	**0.568**	**0.528**	**0.547**
100	0.451	0.198	0.540	0.525	0.532

TABLE X
COMPARISON OF MULTIDIMENSIONAL PHYSICAL PARAMETER CLUSTERING

Multidimensional parameter clustering	mAP50	mAP50:95	P	R	F1
✗	0.463	0.200	0.545	0.508	0.526
✓	**0.468**	**0.205**	**0.568**	**0.528**	**0.547**

because overfitting edge noise destroys topological continuity. When $\lambda_{rec} = 0.4$ and $\lambda_f = 20$, the gradient magnitude achieves perfect alignment, obtaining an optimal mAP50 of 0.468.

As shown in Table IX, the experiment on the foreground weight W further reveals the foreground and background gradient check and balance mechanism. When $W = 1$, the background gradient dominates, and Precision is only 0.512. When $W = 10$, foreground gradient focusing makes the network pay more attention to real physical features, precision rises to 0.568, and mAP50 reaches 0.468. However, when $W = 100$, the excessive weight destroys the classification boundary and triggers massive false alarms, leading to a performance fallback.

4) Comparison of Multidimensional Physical Parameter Clustering: As shown in Table X, this section compares the impact of traditional clustering strategies and the strategy introducing multidimensional physical parameters on detection performance.

Quantitative data confirms that the introduction of physical parameters significantly improves model performance. When only employing traditional clustering, the model mAP50 is 0.463, and precision and recall are 0.545 and 0.508, respectively. At this time, clustering only relies on spatial position and echo intensity, making it easy to mistakenly group structures with vastly different scattering mechanisms into one category, leading to semantic flaws in part masks.

After introducing parameters such as frequency dependence factor, distribution length, and azimuth angle, the model

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performance leaps comprehensively. The mAP50 increases to 0.468, precision significantly increases to 0.568, and recall synchronously rises to 0.528. Its underlying mechanism is that multidimensional parameters inject strong physical constraints into clustering, which can distinguish electromagnetic scattering mechanisms such as edge diffraction and corner reflection, making the clustering boundary fit the true physical structure of targets.

IV. Conclusion

To address the challenges of feature discreteness and scattering voids caused by complex scattering mechanisms in SAR target detection, this article proposes the PRIF. By integrating the ASC part model with deep learning, this framework effectively bridges the gap between pure data-driven paradigms and the underlying electromagnetic scattering mechanisms. Methodologically, the proposed framework introduces three novel core modules. First, the TSCM module resolves the label permutation problem inherent in unsupervised clustering, transforming disordered scattering points into semantically aligned supervisory signals. Second, the PALA strategy rectifies the positive sample selection criterion from the traditional geometric center to physical entity alignment, thereby substantially enhancing the capture of targets exhibiting scattering voids. Finally, the core network, PRI-Net, significantly boosts both part-level local feature discrimination and model interpretability via its PPRB and PDDB branches. Extensive experiments comprehensively validate the effectiveness of the proposed framework. On the FAIR-CSAR dataset, PRIF significantly outperforms existing state-of-the-art algorithms. Furthermore, cross-dataset validation on the RSAR dataset and portability experiments on baseline models demonstrate its robust generalization capabilities and plug-and-play characteristics.

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